Chapter -2 (प्रतिलोम त्रिकोणमितीय फलन)

Important Formula

- $\sin^{-1} x = \theta \Leftrightarrow \sin\theta = x$
- $\cos^{-1} x = \theta \Leftrightarrow \cos\theta = x$
- $\tan^{-1} x = \theta \Leftrightarrow \tan\theta = x$

- $\cos(\cos^{-1} x) = x,$
- $\sin(\sin^{-1} x) = x, -1 \le x \le 1$
- $\tan(\tan^{-1} x) = x, \quad -\infty < x < \infty$
- $\cot(\cot^{-1} x) = x, \quad -\infty < x < \infty$
- $\sec(\sec^{-1} x) = x \quad x \le -1$ या $x \ge 1$
- $\text{cosec}(\text{cosec}^{-1} x) = x\ , x \le -1$ या $x \ge 1$

- $\sin^{-1}(\sin\theta) = \theta$ केवल यदि $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$
- $\cos^{-1}(\cos\theta) = \theta$ केवल यदि $0 \le \theta \le \pi$
- $\tan^{-1}(\tan\theta) = \theta$ केवल यदि $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$
- $\cot^{-1}(\cot\theta) = \theta$ केवल यदि $0 < \theta < \pi$
- $\sec^{-1}(\sec\theta) = \theta$ केवल यदि $0 \le \theta \le \pi, \theta \ne \frac{\pi}{2}$
- $\text{cosec}^{-1}(\text{cosec}\theta) = \theta$ केवल यदि $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}, \theta \ne 0$

- $\sin^{-1} x = \text{cosec}^{-1}\frac{1}{x}, \quad -1 \le x < 0$ या $0 < x \le 1$
- $\cos^{-1} x = \sec^{-1}\frac{1}{x}, \quad -1 \le x < 0$ या $0 < x \le 1$
- $\tan^{-1} x = \cot^{-1}\frac{1}{x}$ केवल यदि $x > 0$ यदि $x < 0$ है तो

Imp

- $\sin^{-1}(-x) = -\sin^{-1} x, -1 \le x \le 1$
- $\cos^{-1}(-x) = \pi - \cos^{-1} x, -1 \le x \le 1$
- $\tan^{-1}(-x) = -\tan^{-1} x, x \in R$
- $\cot^{-1}(-x) = \pi - \cot^{-1} x, x \in R$
- $\sec^{-1}(-x) = \pi - \sec^{-1} x, x \le -1$ या $x \ge 1$
- $\text{cosec}^{-1}(-x) = -\text{cosec}^{-1} x, x \le -1$ या $x \ge 1$

Imp

- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, x \in [-1, 1]$
- $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, x \in R$
- $\sec^{-1} x + \text{cosec}^{-1} x = \frac{\pi}{2}, |x| \ge 1$

Imp

- $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\frac{x+y}{1-xy}$
- $\tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1}\frac{x+y}{1-xy}$

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1}\frac{x+y+z-xyz}{1-xy-yz-zx}$$

- $\cot^{-1}x + \cot^{-1}y = \cot^{-1}\frac{xy-1}{y+x} \quad xy > -1$
- $\cot^{-1}x - \cot^{-1}y = \cot^{-1}\frac{xy+1}{y-x}$

Imp

- $\sin^{-1}x \pm \sin^{-1}y = \sin^{-1}\left[x\sqrt{1-y^2} \pm y\sqrt{1-x^2}\right]$
- $\cos^{-1}x \pm \cos^{-1}y = \cos^{-1}\left[xy \mp \sqrt{1-x^2}\cdot\sqrt{1-y^2}\right]$

Imp

- $2\sin^{-1}x = \sin^{-1}\left[2x\sqrt{1-x^2}\right]$
- $2\cos^{-1}x = \cos^{-1}(2x^2-1)$
- $2\tan^{-1}x = \tan^{-1}\frac{2x}{1-x^2} = \sin^{-1}\frac{2x}{1+x^2} = \cos^{-1}\frac{1-x^2}{1+x^2}$
- $2\cot^{-1}x = \cot^{-1}\frac{x^2-1}{2x}$

Imp

- $3\sin^{-1}x = \sin^{-1}(3x-4x^3)$
- $3\cos^{-1}x = \cos^{-1}(4x^3-3x)$
- $3\tan^{-1}x = \tan^{-1}\frac{3x-x^3}{1-3x^2}$

Imp

- $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2} = \tan^{-1}\frac{x}{\sqrt{1-x^2}} = \cot^{-1}\frac{\sqrt{1-x^2}}{x} = \sec^{-1}\frac{1}{\sqrt{1-x^2}} = \mathrm{cosec}^{-1}\frac{1}{x}$
- $\cos^{-1}x = \sin^{-1}\sqrt{1-x^2} = \tan^{-1}\frac{\sqrt{1-x^2}}{x} = \cot^{-1}\frac{x}{\sqrt{1-x^2}} = \sec^{-1}\frac{1}{x} = \mathrm{cosec}^{-1}\frac{1}{\sqrt{1-x^2}}$
- $\tan^{-1}x = \sin^{-1}\frac{x}{\sqrt{1+x^2}} = \cos^{-1}\frac{1}{\sqrt{1+x^2}} = \cot^{-1}\frac{1}{x} = \sec^{-1}\sqrt{1+x^2} = \mathrm{cosec}^{-1}\frac{\sqrt{1+x^2}}{x}$